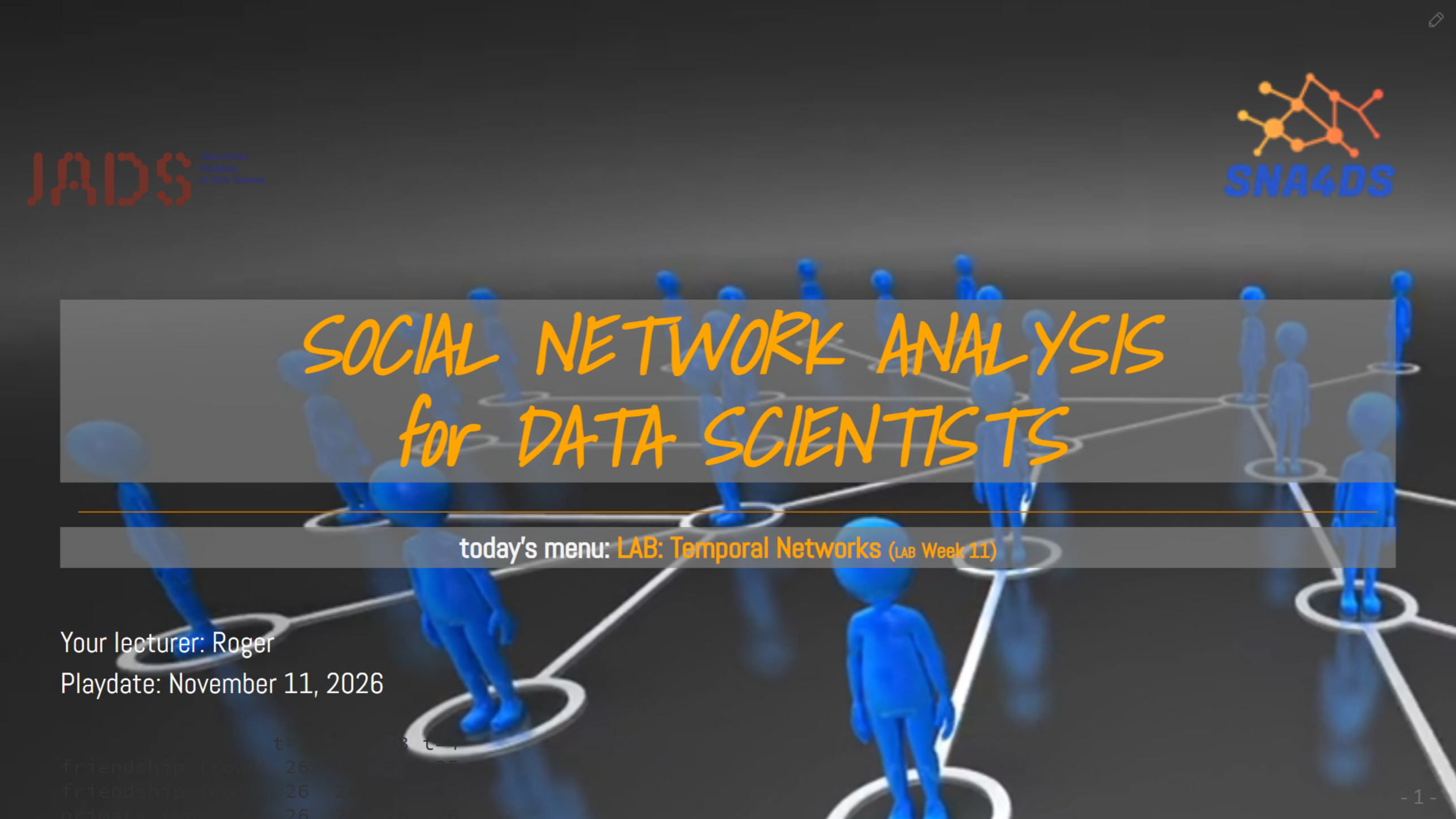


SOCIAL NETWORK ANALYSIS for DATA SCIENTISTS

today's menu: LAB: Temporal Networks (LAB Week 11)

Your lecturer: Roger

Playdate: November 11, 2026



Any SNA4DS-related questions?

Longitudinal classroom friendship network and behavior (Andrea Knecht)

Description

Longitudinal classroom friendship network and behavior (Andrea Knecht).

Format

Note: the data have to be transformed before they can be used with **btergm** and related packages (see examples below).

`friendship`

is a list of adjacency matrices at four time points, containing friendship nominations of the column node by the row node. The following values are used: `0` = no, `1` = yes, `NA` = missing, `10` = not a member of the classroom (structural zero).

`demographics`

is a data frame with 26 rows (the pupils) and four demographic variables about the pupils:

- `sex` (`1` = girl, `2` = boy)
- `age` (in years)
- `ethnicity` (`1` = Dutch, `2` = other, `0` = missing)
- `religion` (`1` = Christian, `2` = non-religious, `3` = non-Christian religion, `0` = missing)

`primary`

is a 26 x 26 matrix indicating whether two pupils attended the same primary school. `0` = no, `1` = yes.

`delinquency`

is a data frame with 26 rows (the pupils) and four columns (the four time steps). It contains the rounded average of four items (stealing, vandalizing, fighting, graffiti).

Homeplay: Model 1

R Code | Summary | Goodness of fit

```
model.01 <- btergm::btergm(friendship ~ edges + mutual + ttriple +
                      transitiveties + ctriple + nodeicov("idegsqrt") +
                      nodeicov("odegsqrt") + nodeocov("odegsqrt") +
                      nodeofactor("sex") + nodeifactor("sex") + nodematch("sex") +
                      edgecov(primary), R = 100)

btergm::summary(model.01)

snafun::stat_plot_gof_as_btergm(model.01,
             btergm_statistics = c(btergm::esp, btergm::dsp, btergm::geodesic,
                                   btergm::deg, btergm::triad.undirected, btergm::rocpr))
```

Homeplay: Model 1

R Code | Summary | Goodness of fit

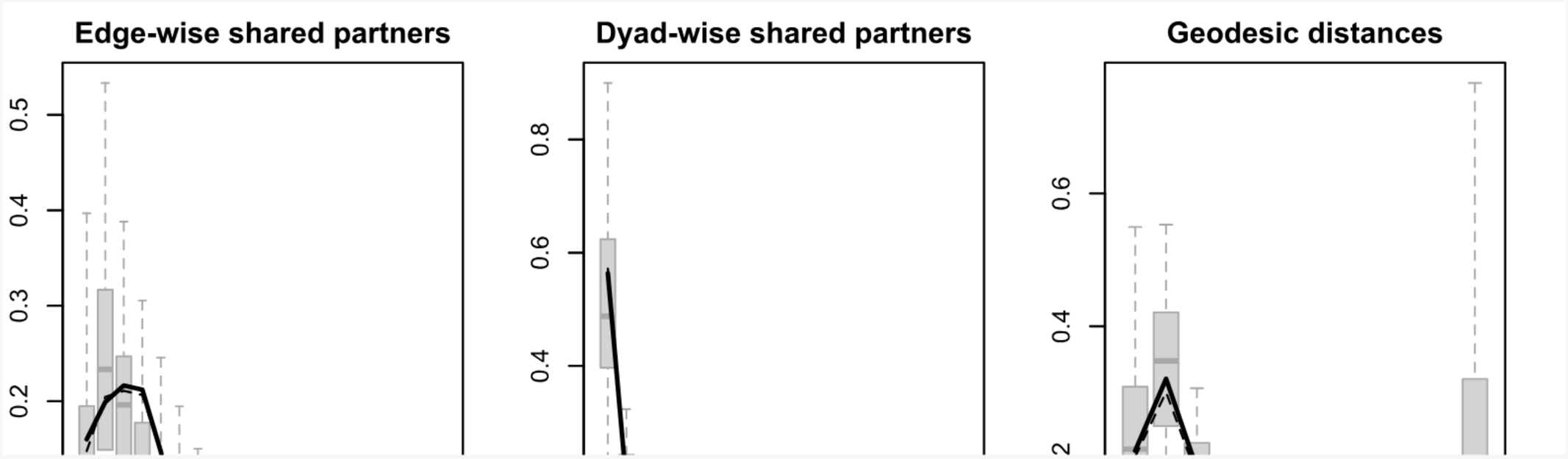
```
btergm::summary(model.01)
```

```
                     Estimate Boot mean     2.5%    97.5%
edges                -9.17591  -9.31582 -10.3377  -8.6755
mutual                2.96172   2.99616   2.2630   3.7861
ttriple               0.21316   0.21734   0.0955   0.3509
transitiveties        0.36794   0.35113   0.2114   0.4292
ctriple              -0.67677  -0.68368  -0.9483  -0.4596
nodeicov.idegsqrt     1.17509   1.22274   0.9788   1.5237
nodeicov.odegsqrt    -0.28432  -0.32635  -0.5821  -0.1682
nodeocov.odegsqrt     1.19846   1.24066   1.1665   1.4090
nodeofactor.sex.2     0.60720   0.62415   0.4228   0.8305
nodeifactor.sex.2     0.22578   0.22499   0.0487   0.3592
nodematch.sex         1.76805   1.78569   1.6030   2.0681
edgecov.primary[[i]]  1.05123   1.01450   0.7702   1.2552
```

Homeplay: Model 1

R Code | Summary | Goodness of fit

```
snafun::stat_plot_gof_as_btergm(model.01,
            btergm_statistics = c(btergm::esp, btergm::dsp, btergm::geodesic,
                                  btergm::deg, btergm::triad.undirected, btergm::rocpr))
```



Intermezzo: rusty on ROC and PR?

ROC: plots FPR (False Positive Rate, x-axis) against TPR (True Positive Rate, y-axis)

TPR = fraction of ties that are correctly predicted by the model

FPR = faction of 0's in the data that are incorrectly predicted as 1

You want TPR to be very high and FPR to be very low. Hence: the curve should lean strongly into the top-left corner of the plot, this indicates how well the model can distinguish between 1's and 0's in the networks.

(the ROC of a poor model will be close to the diagonal)

Best suited for balanced datasets, where the number of 1's and 0's is similar.

PR: plots TPR (True Positive Rate, x-axis) against Precision (y-axis)

Precision = of all predicted 1's, which proportion is correct

You want TPR and Precision to be very high. Hence: the curve should lean strongly into the top-right corner

Best suited for imbalanced datasets — where 1's are rare

(a poor model will just mimic the network's density horizontally)

Model II

R Code | Summary | Goodness of fit

```
model.02 <- btergm::btergm(friendship ~ edges + mutual + ttriple +
                      transitiveties + ctriple + nodeicov("idegsqrt") + nodeicov("odegsqrt") +
                      nodeocov("odegsqrt") + nodeofactor("sex") + nodeifactor("sex") +
                      nodematch("sex") + edgecov(primary) + delrecip +
                      memory(type = "stability"), R = 100)

btergm::summary(model.02)

snafun::stat_plot_gof_as_btergm(model.02,
             btergm_statistics = c(btergm::esp, btergm::dsp, btergm::geodesic,
                                   btergm::deg, btergm::triad.undirected, btergm::rocpr))
```

Model II

R Code | Summary | Goodness of fit

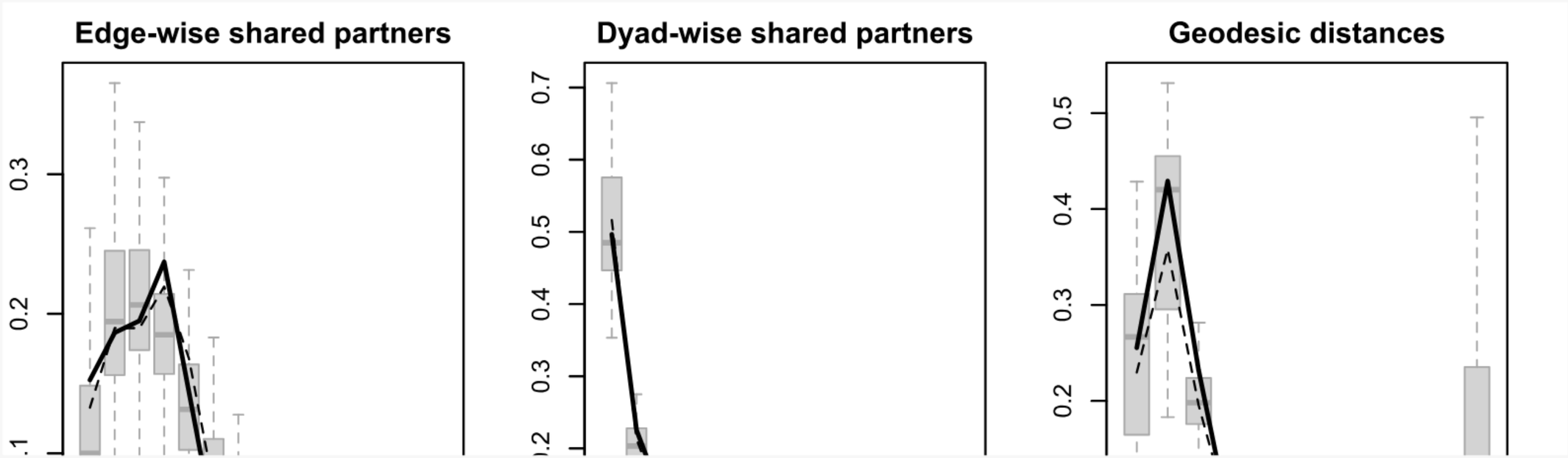
```
btergm::summary(model.02)
```

```
                      Estimate Boot mean     2.5%    97.5%
edges                 -9.54228  -9.78141 -11.4719  -8.7203
mutual                 2.17294   2.19220   1.8416   2.6791
ttriple                0.13423   0.12964   0.0345   0.2123
transitiveties         0.31509   0.32739   0.2884   0.3945
ctriple               -0.55434  -0.56112  -0.7623  -0.4185
nodeicov.idegsqrt      1.28064   1.32425   1.1461   1.5899
nodeicov.odegsqrt     -0.13179  -0.14649  -0.3148  -0.0353
nodeocov.odegsqrt      1.49624   1.54988   1.3781   1.8553
nodeofactor.sex.2      0.52990   0.56336   0.3846   0.8017
nodeifactor.sex.2      0.28694   0.29877  -0.0012   0.5483
nodematch.sex          1.49245   1.52131   1.3044   1.8714
edgecov.primary[[i]]   0.42395   0.41395  -0.0552   0.8300
edgecov.delrecip[[i]]  0.67436   0.73580   0.3309   1.3391
edgecov.memory[[i]]    0.77922   0.79105   0.6961   0.9556
```

Model II

R Code | Summary | Goodness of fit

```
snafun::stat_plot_gof_as_btergm(model.02,
            btergm_statistics = c(btergm::esp, btergm::dsp, btergm::geodesic,
                                  btergm::deg, btergm::triad.undirected, btergm::rocpr))
```



Model III: out-of-sample GOF

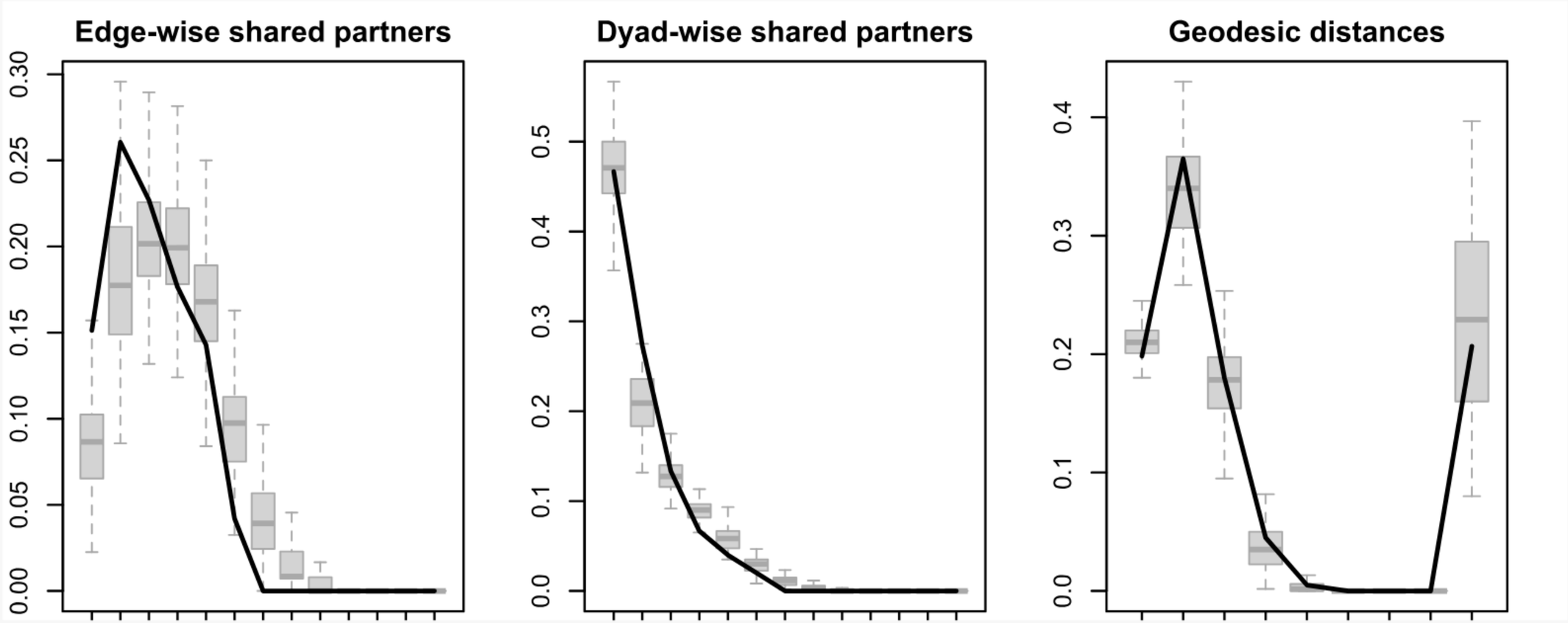
R Code | GOF of out of sample network

```
model.03 <- btergm::btergm(friendship[1:3] ~ edges + mutual + ttriple +
                      transitiveties + ctriple + nodeicov("idegsqrt") + nodeicov("odegsqrt") +
                      nodeocov("odegsqrt") + nodeofactor("sex") + nodeifactor("sex") +
                      nodematch("sex") + edgecov(primary) + delrecip() +
                      memory(type = "stability"), R = 100)

# out of sample gof
gof.03 <- btergm::gof(model.03, nsim = 100, target = friendship[[4]],
               formula = friendship[3:4] ~ edges + mutual + ttriple +
                 transitiveties + ctriple + nodeicov("idegsqrt") +
                 nodeicov("odegsqrt") + nodeocov("odegsqrt") + nodeofactor("sex") +
                 nodeifactor("sex") + nodematch("sex") + edgecov(primary) +
                 delrecip + memory(type = "stability"),
               statistics = c(btergm::esp, btergm::dsp, btergm::geodesic,
                              btergm::deg, btergm::triad.undirected, btergm::rocpr))
```

Model III: out-of-sample GOF

R Code | GOF of out of sample network



The probability of an edge



Get all probabilities using `btergm::edgeprob(model.03)`

Extract what you need:

```
ep[ep$i == 10 & ep$j == 15 & ep$t == 2, "probability"]
```

```
[1] 0.8243134
```

```
ep[ep$i == 10 & ep$j %in% c(9, 11, 12, 15) & ep$t == 2, "probability"]
```

```
[1] 0.04576358 0.22475778 0.30616613 0.82431338
```

Note: you can also use `btergm::interpret`.

Edge probability for specific cases

R Code | Summary

```
model.04 <- btergm::btergm(friendship ~ edges + mutual + nodematch("sex") +
                      edgecov(primary), R = 100)

btergm::summary(model.04)
```

(terrible model, but that doesn't matter for now)

Edge probability for specific cases

R Code | Summary

```
btergm::summary(model.04)
```

```
                     Estimate Boot mean    2.5%   97.5%
edges                 -3.0875   -3.1269 -3.7262 -2.7424
mutual                 1.9971    2.0017  1.7416  2.2753
nodematch.sex          1.1957    1.2240  1.0691  1.5008
edgecov.primary[[i]]   1.2733    1.2958  1.0898  1.5953
```

R Code | Summary

What is the probability of an edge between *i* and *j* if

- there is no tie from *j* to *j*
- the actors have a different gender
- attended a different primary school

```
btergm::summary(model.04)
```

```
                     Estimate Boot mean    2.5%   97.5%
edges                 -3.0875   -3.1269 -3.7262 -2.7424
mutual                 1.9971    2.0017  1.7416  2.2753
nodematch.sex          1.1957    1.2240  1.0691  1.5008
edgecov.primary[[i]]   1.2733    1.2958  1.0898  1.5953
```

R Code | Summary

In this case *mutual* = *nodematch.sex* = *edgecov[i]* = 0. So, the logodds of this tie is -3.0875.

```
1 / (1 + exp(+3.0875))
```

```
[1] 0.04362582
```

```
plogis(-3.0875)
```

```
[1] 0.04362582
```

Note: this is the overall probability for this tie across the four waves.

R Code | Summary

What is the probability of an edge between *i* and *j* if

- there is a tie from *j* to *j*
- the actors have a different gender
- attended the same primary school

```
btergm::summary(model.04)
```

```
                     Estimate Boot mean    2.5%   97.5%
edges                 -3.0875   -3.1269 -3.7262 -2.7424
mutual                 1.9971    2.0017  1.7416  2.2753
nodematch.sex          1.1957    1.2240  1.0691  1.5008
edgecov.primary[[i]]   1.2733    1.2958  1.0898  1.5953
```

R Code | Summary

The logodds of the tie is now -3.0875 + 1.9971 + 1.2733 = 0.1829.

```
plogis(0.1829)
```

```
[1] 0.545598
```

Note: this is the overall probability for this tie across the four waves.

ANY QUESTIONS?